Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 8

Deadline: by the end of next Wednesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems. **Justify your answers.**

Problem 1. Give an example of a group G and elements a, b, and c in G, such that a is different from b but $ac = cb$.

Problem 2. Let G be a cyclic group generated by x. Note that for any positive integer k, the set $\langle x^k \rangle$ is a subgroup of G. If x has finite order n, consider the possible values for k (from 0 to $n-1$) and, in each case, show that $\langle x^k \rangle$ is generated by x^d where d is the greatest common divisor of k and n. Deduce that $\langle x^k \rangle$ has n/d elements.

Problem 3. Let G be a cyclic group generated by x with x of order greater than 1. Let H be a subgroup of G with H neither G nor $\{e\}$.Le $m \geq 1$ be minimal such that $x^m \in H$. Use the division algorithm to show that x^m generates H and deduce that every subgroup of a cyclic group is cyclic.

Problem 4. Let g and x be elements of a group G. Show that for all positive integers k, we have $(g^{-1}xg)^k = g^{-1}x^kg$. Deduce that x has order 3 if and only if (for all $g \in G$) $g^{-1}xg$ has order 3. Show that the same is true for any integer $n \geq 1$ in place of 3.